# Citation check for proposal version 10

Each row pairs a claim in `docs/proposal/proposal_v10.md` with the exact words of the source it cites. The quotes come from the saved copies in `data/external/` and from our own reports. Page numbers are the PDF page, with the printed page where it differs. The fit column says whether the proposal states the source exactly, paraphrases it closely, or rests on our own work.

Version 10 cites the same 18 sources as versions 4 to 9 and adds no new outside source. Its citations are renumbered in order of first use, so the integrity report is now [8], the Award Data Archive [9] and the HHI page [10]. Citations [8] and [13] link to the PDFs in the team repository. The 22 outside quotes are the ones checked word for word against their saved copies on 2026-10-07 for version 4. Every quote from our own reports was checked word for word against the repository copies on 2026-10-07.

Versions 6 to 10 call one agency, product code and fiscal year a segment. Our integrity report [8] calls the same unit a cell, so its quotes keep that word.

## Outside sources

| Claim in the proposal | Source | Direct quote | Fit |
|---|---|---|---|
| The government obligated $793 billion through contracts in FY2025. | [1] GAO, FY2025 snapshot data file, field "intro" | "The federal government obligated $793 billion through contracts in fiscal year 2025, a $17.8 billion increase from fiscal year 2024 after adjusting for inflation." | Exact |
| US aerospace and defense prime contractors fell from 51 in 1993 to 5 in 2000. | [2] DoD, State of Competition within the Defense Industrial Base, p. 23 (PDF p. 25) and p. 1 (PDF p. 3) | "Major consolidations in the mid-1990s significantly reduced competition for weapons programs, with the total number of U.S.-based prime contractors declining from 51 in 1993 to 5 in 2000 [Figure 2]." Also: "Since the 1990s, the defense sector has consolidated substantially, transitioning from 51 to 5 aerospace and defense prime contractors." | Exact. The dates come from p. 23 and the "aerospace and defense" wording from p. 1. |
| In defense, a few firms now lead most major weapon programs. At least one of those five is a prime contractor on 58 of the 78 major weapon systems in DoD's FY2024 budget. | [3] CRS R47751, p. 9 (PDF p. 13) | "Of the 78 major weapons systems DOD identified in its FY2024 budget materials, 58—over 74% of the total—feature at least one of the ‘Big Five’ as a prime contractor." | Exact for 58 of 78. "Most major weapon programs" is our plain summary of that 74%. |
| Studies that measure concentration directly also stay within defense. | [4] Carril and Duggan, abstract, p. 1 | "We focus on the defense industry, by far the largest contributor to federal procurement spending in the U.S." Also: "Using detailed microdata on Department of Defense (DoD) contract awards, we estimate the causal effect of industry concentration on a series of procurement outcomes" | Exact |
| Same claim | [5] CSIS, Evaluating Consolidation and the Threat of Monopolies, abstract, PDF p. 3 | "This paper examines the importance of industrial concentration in two ways: first, a direct relationship between concentration and performance outcomes; and second, an indirect relationship, where concentration influences performance through reduced competition for defense acquisition." | Close. The study measures outcomes of defense acquisition. |
| Same claim | [6] CSIS, Defense Acquisition Trends 2023, PDF p. 10 | "since FY 2000 they have accounted for at least 27.8 percent of prime contract obligations. The Big Five’s share of DoD obligations decreased in FY 2021 to 29.4 percent from an FY 2020 high of 36.0 percent." | Close. It measures the five largest firms' share of DoD obligations only. |

| Claim in the proposal | Source | Direct quote | Fit |
|---|---|---|---|
| The closest study compares DoD with all other agencies combined but measures concentration for only five product categories, over FY2010 to FY2019. | [7] Hyatt, Acquisition Research Symposium 2023, PDF p. 7 and p. 9 | "the effective competition, noneffective competition, and non-competed contract actions for all products and services were calculated for the DoD and all other federal agencies combined." Also: "Due to page constraints, only FY10–FY19 results are shown and discussed below" Also: "These five PSCs cumulatively represented roughly 12%–15% of the total dollars obligated by the federal government each year from FY10–FY19." | Exact. Hyatt's five PSCs are product and service codes, which the summary calls product categories. |
| We use every federal contract transaction in USAspending's Award Data Archive. | [9] USAspending, Award Data Archive page | "Download major agencies' award transaction data for full fiscal years." | The source of the data. The row count, 88,753,968, is our own count in [8]. |
| HHI is the sum of each vendor's squared percentage share and runs from near 0 to 10,000. | [10] DOJ, Herfindahl-Hirschman Index page | "The HHI is calculated by squaring the market share of each firm competing in the market and then summing the resulting numbers." Also: "It approaches zero when a market is occupied by a large number of firms of relatively equal size and reaches its maximum of 10,000 points when a market is controlled by a single firm." | Exact |
| Product codes changed meaning in FY2012. | [11] GSA, PSC Manual, August 2011, PDF p. 10 | "384 ended codes were replaced with 504 new codes (captured new and/or expanded numbering scheme for facility codes)" | Exact. The manual took effect on October 1, 2011, the first day of FY2012. |
| Product codes changed meaning again in FY2021. | [12] GSA, PSC Manual, October 2020, PDF p. 8 | "One Hundred Fifty-Five (155) Product Service Codes (PSCs) were added. These new R&D codes have a scheme aligned to OMB Budget Function, and will be activated October 30, 2020." Also: "Seven Hundred twenty-One (721) R&D PSCs are being end-dated on October 29, 2020 by new R&D codes starting with "A"." | Exact. October 30, 2020 falls in FY2021. |
| Our yearly totals match GAO within 0.2% for FY2022 to FY2025. | [14], [15], [16] and [1], GAO snapshot data files, field "intro" | FY2022: "The federal government obligated $694 billion through contracts in fiscal year 2022, including about $40 billion related to the COVID-19 pandemic." FY2023: "The federal government obligated $759 billion through contracts in fiscal year 2023, a $33 billion increase from fiscal year 2022 after adjusting for inflation." FY2024: "The federal government obligated $755 billion through contracts in fiscal year 2024, a $22.5 billion decrease from fiscal year 2023 after adjusting for inflation." FY2025 is in the first row. | Exact for GAO's figures. The comparison is ours, in [8], with gaps of +0.10%, +0.19%, +0.18% and +0.02%. |
| Cells with HHI above 1,800 are highly concentrated under the 2023 guidelines. | [17] DOJ and FTC, Merger Guidelines, 2023, p. 5 (PDF p. 6) | "Markets with an HHI greater than 1,800 are highly concentrated, and a change of more than 100 points is a significant increase." | Exact |
| Cells with HHI above 2,500 are highly concentrated under the 2010 guidelines. | [18] DOJ and FTC, Horizontal Merger Guidelines, 2010, Sec. 5.3, p. 19 (PDF p. 22) | "Highly Concentrated Markets: HHI above 2500" | Exact |

## Our own reports

| Claim in the proposal | Source | Direct quote | Fit |
|---|---|---|---|
| 88,753,968 rows from FY2008 to FY2025 | [8] Integrity report v3, opening line | "All 88,753,968 federal contract transactions in USAspending's Award Data Archive, release of 2026-09-06, checked without changing any value." | Our work |
| A verified Parquet copy, and 97 CSV files copied without loss | [8], "How the data was checked" | "The 18 yearly zip files hold 97 CSV files. For each one, two independent readers agree on the record count, the exact dollar total and a checksum over every field of every row. All 18 years pass." | Our work |
| Flags from 62 rule checks | [8], "How the data was checked" | "62 checks ran on every row of every year, each with a fixed rule and threshold." | Our work |
| Nine main data decisions | [8], "Summary" | "Each issue comes with a proposed decision for the team, and the nine decisions come first, in one table." | Our work |
| About 365,000 segments | [8], "Negative vendor totals" | "They sit in 127,992 of 365,402 cells (35.0%)" | Our work |
| Five issues would distort the measure. | [8], "Summary" | "Five issues would distort it if the data were used as it stands." | Our work |
| About 7.5% of vendor totals within a segment are negative. | [8], "Summary" | "About 7.5% of vendor totals within a cell are negative." | Our work |
| Totals within 0.2% of GAO for FY2022 to FY2025 | [8], "Summary" | "Our yearly totals match GAO's figures within 0.2% in the four years GAO publishes exact figures, FY2022 to FY2025." | Our work |
| Parent company IDs can lag a merger by 2 to 6 months. New in version 5. | [8], "Which company a row belongs to" | "The switch comes 2 to 6 months after each closing, and a small share of rows keeps the old ID for months after that." | Our work. It rests on four deals from 2013 to 2020. FY2008 to FY2012 are untested. |
| Product codes changed meaning in FY2021. | [13] Research note v3, section 1 | "139 of the 155 new R&D codes reuse an old code string with a new meaning. AC11 meant defense aircraft basic research until October 29, 2020. From the next day it means all Department of Defense military basic research (E02)." | Our work, drawn from [12] and GSA's code list |

## Claims with no citation

These rest on our own computations or reasoning, so the proposal does not cite a source for them. Each traces to a file in the repository or to plain arithmetic.

- **Research on concentration has focused on defense.** This sums up the studies we found, [4] to [7]. Our search was not exhaustive, so the proposal says "we found no measure" for the rest of the government.
- **Each transaction carries a product or service code, a four-character label, such as 1510 for fixed-wing aircraft.** Check F01 in `config/flag_catalog.csv` tests that every product code has four characters. The label for 1510 appears in [8], in the table under "Large entries that were reversed later" ("Defense, 1510 fixed-wing aircraft, FY2020").
- **The measure counts prime contractors only.** The archive files are prime transaction files. `reports/archive_vs_custom_export_q1fy2025.md` matched the archive's FY2025 rows against USAspending's own download for Q1 FY2025, whose files are named `All_Contracts_PrimeTransactions`. Subcontracts sit in separate subaward files that we do not use.
- **The measure looks at one buyer at a time, so the antitrust cutoffs are only a rough guide.** This is our reasoning. The merger guidelines [17], [18] judge concentration across a market of all buyers. Our segments cover one agency's purchases, and a vendor that wins most of them can still face many rivals for other buyers.
- **DuckDB lets one laptop query all 88.8 million rows.** Stages 1 and 2 of the pipeline ran this way on one laptop.
- **The preliminary table and the 30% of single-vendor segments** come from `reports/proposal/concentration_prelim_by_year.csv` and `concentration_prelim_hhi_bands.csv`.

- **The $71 billion of the $130 billion rise, and the fixed-wing aircraft figures,** come from `reports/proposal/concentration_prelim_hhi_bands.csv` and `concentration_prelim_largest_cells.csv`.
- **Offers received coded consistently only from FY2012 to FY2017** comes from [8], "Some fields fill in or empty out over time". It says "FY2011 still has 64,905 competed actions with zero offers, against 14,557 or fewer in each year from FY2012, so FY2012 to FY2017 is the cleaner window."
- **HHI needs no inflation adjustment.** Inflation scales every vendor in a segment by the same factor. Every share stays the same, and so does the HHI.
- **The examples of 2,000 and 6,500** are arithmetic. Both agencies spend $100M on one product. Five vendors at $20M each hold 20% apiece, and $20^2 \times 5 = 2{,}000$. One vendor at $80M holds 80% and four at $5M hold 5% each, and $80^2 + 5^2 \times 4 = 6{,}500$. The same example appears in our team pitch.
- **The rising rule.** A line is rising if its average HHI over its last 3 years is at least 500 points above its average over its first 3 years, with runs at 250 and 1,000 points. This is our choice, explained in `docs/proposal/proposal_v10_methods_notes.md`, section 2. No source fixes the cutoff.
- **Sole-source share measured for umbrella contracts and for the orders under them.** This follows the plan in our team pitch, "Measure both ways, at the job level and the umbrella level, and say which is which." The fields and codes are in the methods notes, section 3, and still need checking against the FPDS data dictionary.
- **Features from each line's first 3 years and the starting vendor count.** This is our reasoning. A falling vendor count raises HHI by itself.
- **Competed awards with a single offer, for FY2012 to FY2017 for now.** The window comes from [8], quoted in the item on offers received.
- **Our total is within 2% of GAO's in every year GAO reports one.** [8] says GAO publishes exact figures for FY2022 to FY2025, "Our yearly totals match GAO's figures within 0.2% in the four years GAO publishes exact figures, FY2022 to FY2025."
- **Our own contribution.** The measure, the pipeline and the review of the data's gaps are this team's work, in this repository.
- **A code whose meaning changed starts a new trend line.** This is our rule for the two product code changes, [11] and [12].
- **The note under the preliminary table.** Segments with no positive-net vendor, 2,282 in FY2024 and 2,868 in FY2025, and negative vendor totals of 9.2% and 9.9% in those years, come from the first independent review of the repository, `docs/reviews/2026-10-07_github/summary.md`, rows 3 and 4, which checked them against `src/concentration_prelim.py` and its outputs.
- **The AUC rule.** A gain of at least 0.05 with a 95% bootstrap interval above zero is the team's choice. No source fixes the margin.
- **The analysis methods** (k-means clustering, average silhouette width, L1-penalized logistic regression, a shallow decision tree, the adjusted Rand index and cross-validated AUC) are standard methods with no citation in this version. Each could take a textbook or original-paper reference once a saved copy can be checked word for word.